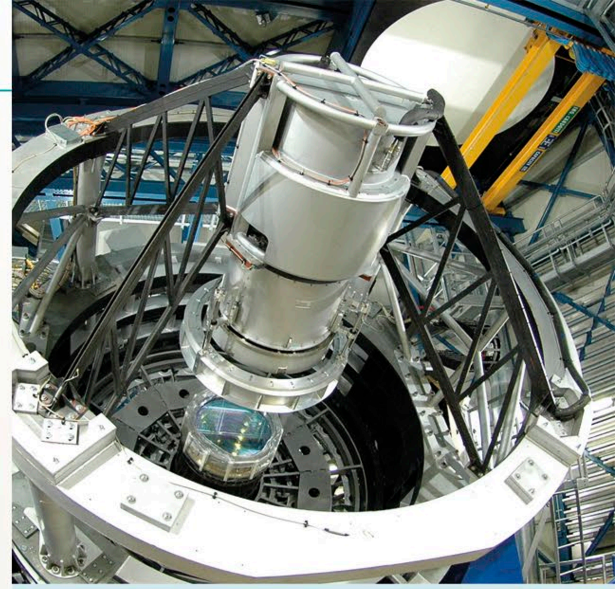
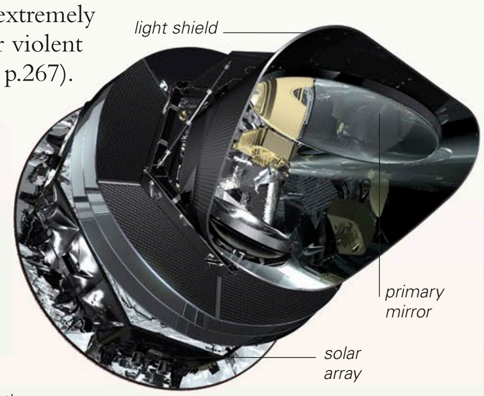
ACROSS THE SPECTRUM

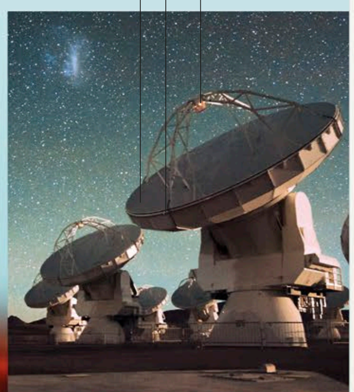
Celestial objects can emit radiation across the EM spectrum, from radio waves through visible light to gamma rays. Some complex objects, such as galaxies and supernova remnants, shine at nearly all these wavelengths. Cool objects tend to radiate photons with less energy and are therefore only visible at longer wavelengths. Toward the gamma-ray end of the spectrum, photons are increasingly powerful. High-energy X-rays and gamma rays originate only from extremely hot sources, such as the gas of galaxy clusters (see p.327) or violent events, such as the swallowing of matter by black holes (see p.267). To detect all this radiation and form images, astronomers need a range of instruments—each type of radiation has different properties and must be collected and focused in a particular way. Radiation at many wavelengths does not penetrate to Earth's surface and is detectable only by orbiting observatories above the atmosphere.





RADIO WAVES TELESCOPE ARRAY

Radio waves can be many feet (meters) long. To create sharp images from such long waves, astronomers collect and focus them using telescopes with huge dish antennae. They might use a single dish or an entire array. The Atacama Large Millimeter Array, or ALMA (right), in northern Chile is the world's largest array. It consists of 66 high-precision antennas that can be moved across a desert plateau into various configurations and their data combined to form a single, fine-detailed image.

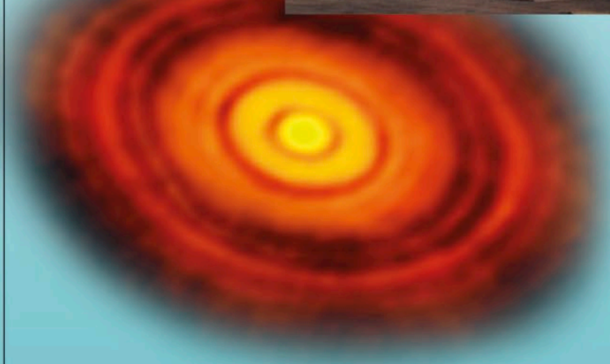


MICROWAVES SPACE OBSERVATORY

Most microwaves are absorbed by Earth's atmosphere, so microwave observatories must be placed in space. Between 2009 and 2013, the European Space Agency's Planck Observatory (above) mapped the cosmic microwave background radiation (see p.54) across the whole sky. This is the oldest EM radiation in the universe, released soon after the Big Bang. The observatory carried out its survey from a stable orbit around the Sun 900,000 miles (1.5 million km) from Earth.

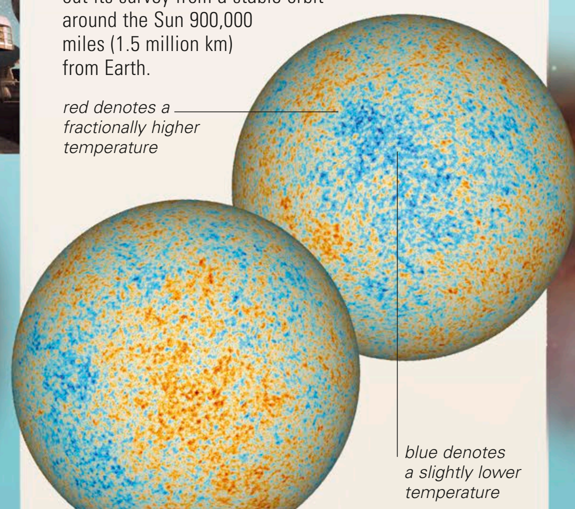
INFRARED MOUNTAINTOP TELESCOPE

Little infrared radiation from space reaches sea level on Earth, but some penetrates down to the height of mountaintops. Some infrared telescopes, such as NASA's Spitzer Space Telescope (see p.247), have been launched into space, but most infrared astronomy is conducted from mountain observatories. This one, named VISTA (Visible and Infrared Survey Telescope for Astronomy), is located at the Paranal Observatory in northern Chile at an altitude of 8,645 ft (2,635 m). It is the world's largest telescope dedicated to surveying in the near infrared (the band of infrared wavelengths closest to visible light). With a 13.5-ft (4.1-m) main mirror, VISTA achieves great resolution. It can be used for detecting, imaging, or peering into a variety of celestial objects, from dim galaxies and galaxy clusters to brown dwarfs and dusty nebulae.



RADIO WAVES PROTOPLANETARY DISK

This image produced by ALMA (see above) shows a protoplanetary disk (disk of dust and gas) rotating around a young, distant star named HL Tauri. The dark rings may signify that planets are forming in the disk. To create such an image, a radio dish, or an array, scans a tiny area of sky, gradually building an image by recording variations in radio intensity. Other objects that produce radio emissions include hydrogen clouds in galaxies or (from the synchrotron radiation they emit) active galaxies (see p.320) and pulsars (see p.267).



MICROWAVES UNIVERSE

The lack of microwave sources in the nearby universe is fortunate, as it makes it easier to observe the cosmic background radiation, which reaches Earth at microwave wavelengths. The pattern of microwaves from the whole sky, as measured by the Planck Observatory, is here projected onto two hemispheres.

INFRARED
NEBULA

This infrared image of the Orion Nebula—a bright, star-forming emission nebula—was taken with the VISTA telescope. VISTA's infrared vision means that it can peer deep into the normally hidden dusty regions of the nebula and reveal the many active young stars buried there.